

ALDRIN JOAN PANDIAN W

aldrinjoan6@gmail.com | +91 82208 63200 | linkedin.com/in/aldrin-joan-pandian-w-08215028a | github.com/Aldrin-Joan
aldrin.23bai11298@vitbhopal.ac.in

TECHNICAL SKILLS

Programming Languages: Python, Java, C++

Generative AI & LLMs: Prompt Engineering, LLM Application Development, RAG, Multi-Agent Systems, LLM Fine-Tuning

AI/ML Frameworks & Tools: LangChain, LangGraph, Hugging Face, TensorFlow/Keras, Scikit-learn, YOLO

Cloud & DevOps: Microsoft Azure, FastAPI, Docker, MCP, Git, GitHub, CI/CD

Web & Mobile Development: React, Flutter, React Native, Firebase, Supabase

Enterprise Integrations: Salesforce, ServiceNow

Certifications: AI Engineering – IBM; Certified Ethical Hacker (CEH) v12 – EC-Council (2023–2027)

PROFESSIONAL EXPERIENCE

AI Engineer — Docu3C (Jul 2025 – August 2026)

Seattle, Washington, USA (Remote) · AI Engineer Intern

- Designed and optimized LLM prompt engineering workflows across GPT-5, Claude, and DeepSeek, improving AI response accuracy, contextual understanding, and reliability for production applications.
- Architected and deployed multi-agent AI systems using LangChain, LangGraph, LangSmith, LiteLLM, and Temporal, implementing agent orchestration, tool calling, and autonomous multi-step workflows.
- Built and shipped cloud-native AI applications on Microsoft Azure, integrating Salesforce, ServiceNow, Model Context Protocol (MCP) servers, REST APIs, and automated document-processing pipelines as sole developer.
- Engineered hybrid AI reasoning systems combining deterministic rule engines with LLM agents; developed benchmarking and calibration frameworks to validate AI outputs against expert-labeled ground-truth datasets.
- Led applied research and rapid prototyping in Generative AI, NLP, semantic search, and vector search, driving improvements in production reliability through systematic debugging, monitoring, and performance tuning.
- Owned the full AI development lifecycle end to end — architecture, implementation, testing, deployment, and production support — for multi-agent workflow systems.

Backend & Flutter Developer — amasQIS.ai (Apr 2025 – Oct 2025)

Omen (Remote)

- Contributed to the Front-End team on a Personal Protective Equipment (PPE) Compliance project, building responsive UI components and webpage layouts.
- Served as Backend Developer on a Predictive Maintenance project, integrating a YOLO computer-vision model for automated defect detection.
- Acted as Lead Flutter Developer for two AI-assisted products — an LLM Workspace Adviser and a VLM Workspace Adviser — managing end-to-end mobile application development.

Software Development Intern — ApexPlanet Software Pvt. Ltd. (Remote) (Oct 2024 – Dec 2024)

- Built a dynamic to-do list, an image gallery, and a full E-commerce website as capstone projects, applying responsive design, JavaScript interactivity, API integration, and localStorage.
- Implemented real-time data handling and performance optimization techniques, strengthening core front-end engineering skills.

FEATURED AI & SOFTWARE PROJECTS

LexAI — AI-Powered Legal Technology Platform

- Built a full-stack AI legal assistant using Flutter, React, and FastAPI, implementing hybrid retrieval-augmented generation (RAG), REST APIs, WebSockets, and voice-enabled legal Q&A across web and mobile.

- Deployed the web app on Firebase Hosting and containerized the backend for Google Cloud Run within a monorepo architecture spanning mobile, web, service, and research layers.
- <https://lexai-3fd1a.web.app>

paperstack — MCP Backend for arXiv Research (477+ PyPI downloads)

- Built and published a production-grade MCP server (paperstack-mcp) on PyPI for AI research assistants, implementing arXiv search, semantic indexing, citation analysis, and LLM context generation.
- Delivered asynchronous processing, caching, and REST-based integrations, driving 477+ package downloads via CLI, API, and MCP integration for VS Code and agent frameworks.
- <https://pypi.org/project/paperstack-mcp>

GraviGauge — VS Code Extension for AI Quota Monitoring (1,142+ downloads)

- Built and published a production-ready VS Code extension (TypeScript, Node.js, VS Code API) for real-time AI model quota monitoring, achieving 1,142+ downloads on the Open VSX Registry.
- Implemented REST API integration, session-based authentication, automatic instance detection, and optimized background polling for low-overhead monitoring.
- <https://open-vsx.org/extension/AldrinJoanPandianW/gravigauge>

Bollywood Bias Buster — AI Bias Analysis Platform

- Built an AI-powered bias-detection platform using NLP, LLMs (Mistral-7B), and multimodal vision models (LLaVA-1.5) to identify and report gender bias in movie scripts, posters, and trailers.
- Implemented automated PDF/CSV reporting and data visualization with Python, Streamlit, PyMuPDF, Pandas, and NetworkX.
- https://github.com/Aldrin-Joan/Bolly_Bias_Streamlit_By_Aldrin_Joan_Pandian.git

Quantane — Vehicle Analytics & Trip Intelligence Platform

- Developed an offline-first Flutter app for fuel tracking, GPS trip analytics, and route visualization, integrating Firebase, Supabase, Geolocator, and fl_chart for real-time vehicle performance insights.
- <https://github.com/Aldrin-Joan/QUANTANE.git>

EDUCATION

B.Tech, Computer Science & Engineering (AI & ML) — VIT Bhopal University	2023 – Present	CGPA: 8.81 / 10
Class XII— Carmel Garden Matric Hr. Sec. School, Coimbatore	2023	75.0%
Class X — Carmel Garden Matric Hr. Sec. School, Coimbatore	2021	100%

PUBLICATIONS & RESEARCH

- Pandian, W. Aldrin Joan, et al. "The Role of Artificial Intelligence in 6G Networks, Architecture, Protocol, Transmission, and Applications." RFID, Microwave Circuit, and Wireless Power Transfer Enabling 5/6G Communication. IGI Global Scientific Publishing, 2025. 115-154.
- Pandian, W. Aldrin Joan, I. Jasmine Selvakumari Jeya, and D. Lakshmi. "for Smart Grid Components: Architecture, Applications, Challenges, and Opportunities." Renewable Power for Sustainable Growth: Proceedings of ICRP 2024, Volume 2 2 (2026): 405.
- Aldrin Joan Pandian, W., et al. "AI-Driven Predictive Maintenance for Smart Grid Components: Architecture, Applications, Challenges, and Opportunities." International Conference on Renewable Power. Singapore: Springer Nature Singapore, 2024.
- J. S. Jeya I, A. J. Pandian W and P. K. Mishra, "A Next-Generation Computing Approach for Early and Accurate Lung Cancer Detection Using an Integrated Hybrid Deep Learning Framework," 2025 International Conference on Next Generation Computing Systems (ICNGCS), Coimbatore, India, 2025, pp. 1-8, doi: 10.1109/ICNGCS64900.2025.11183343.
- Jeya, Jasmine Selvakumari, J. Manikandan, and A. Sherin. "Optimizing Video Compression for Real-Time and Bandwidth-Constrained Applications in IoT Enabled Cyber-Physical Systems." 2025 3rd International Conference on Intelligent Cyber Physical Systems and Internet of Things (ICoICI). IEEE, 2025.
- J. S. Jeya I, N. A. More, B. Das, R. Gupta, V. K. Mishra and A. J. Pandian W, "IoT Integrated Smart Medication System using Machine Learning for Enhanced Rural Healthcare," 2025 6th International Conference on IoT Based Control Networks and Intelligent Systems (ICICNIS), Bengaluru, India, 2025, pp. 442-448, doi: 10.1109/ICICNIS66685.2025.11315818.

- Kumar, S., Dhanotiya, A., Jeya, J. S., Sinha, T. K., & Gupta, A. (2026, February). Accuracy-Efficiency Trade-Off Analysis of Deep Learning Models for Breast Cancer Diagnosis Using Ultrasound Images. In 2026 4th International Conference on Intelligent Data Communication Technologies and Internet of Things (IDCIoT) (pp. 1268-1275). IEEE.

ACHIEVEMENTS & LEADERSHIP

- Presented original research at three international conferences: ICRP 2024, ICoICI 2025, and ICNGCS 2025.
- Selected as **UGC NEP SAARTHI Student Ambassador**, representing the National Education Policy initiative on campus (Jan 2025 – Present).
- Attended the CSIR-sponsored National Seminar on Vehicular Cloud Model (2024).

LANGUAGES

English, Tamil, Hindi, Japanese